

Balun Basics and Practical Performance Parameters

Learn about the balun, a special type of transformer used in mixers, amplifiers, and signaling.

Technical Article January 21, 2024 by [Dr. Steve Arar](#)

Baluns, devices that convert between balanced and unbalanced signals, were originally designed to drive differential antennas used in television transmitting systems. Applications of baluns have since extended to include balanced mixers, amplifiers, and signal lines of all types. Despite their widespread use, however, beginners may find the available information about baluns fragmented and confusing. This article aims to provide an overview of how baluns work, along with some of their most important performance parameters and applications.

The Ideal Balun

Electrical signal transfer always requires two conductors. Single-ended (unbalanced) systems carry the signal on one conductor and use the second conductor as a ground. Differential (balanced) systems use both conductors to carry signals that are 180 degrees out of phase with each other.

The component used to interface between balanced and unbalanced configurations is called a *balun*—short for BALAnced-to-UNbalanced. A balun acts as a power splitter, producing two outputs that are equal in magnitude but 180 degrees out of phase with each other.

A balun is a three-port device. One port is unbalanced, while the other two ports work together to form a single, balanced port. Figure 1 illustrates the typical input and output waveforms of an ideal balun in which Port 1 is the unbalanced port, and Ports 2 and 3 form the balanced port.

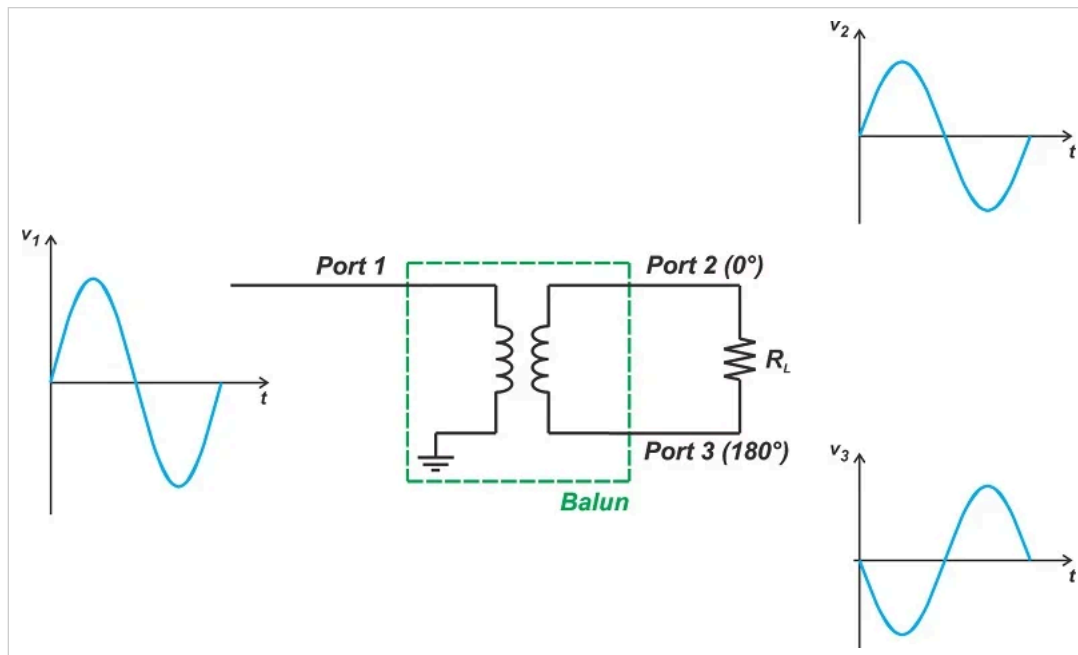


Figure 1. An ideal balun splits the input signal into two signals of equal amplitude but opposite polarity. Image used courtesy of Steve Arar

The following two equations can be used to describe the basic functionality of the balun in terms of its conventional S-parameters. First, we have:

$$S_{21} = -S_{31}$$

Equation 1.

Baluns are reciprocal devices, meaning that they have the same transmission characteristics in both directions. Therefore, in addition to Equation 1, we have:

$$S_{21} = S_{12} = -S_{31} = -S_{13}$$

Equation 2.

Notice that there's no constraint on S_{23} , the transmission between Ports 2 and 3. In other words, the two outputs forming the balanced port might or might not have isolation.

Now that we're familiar with the properties of an ideal balun, let's look at some of the device's most important performance parameters. These include:

- Insertion loss.
- Return loss.
- Amplitude imbalance.
- Phase imbalance.
- Common-mode gain.
- Common-mode rejection ratio.

Insertion Loss

The insertion loss of a balun is also referred to as its differential-mode gain (G_{dm}). With the conventional S-parameters, this parameter is given by:

$$G_{dm} = \frac{S_{21} - S_{31}}{\sqrt{2}}$$

Equation 3.

Balun datasheets will provide values for the single-ended insertion loss at one or more specific frequencies. They may also include curves of S_{21} and S_{31} against frequency, as Figure 2—reproduced from the datasheet of Hyperlabs' [HL9492](#) balun—illustrates.

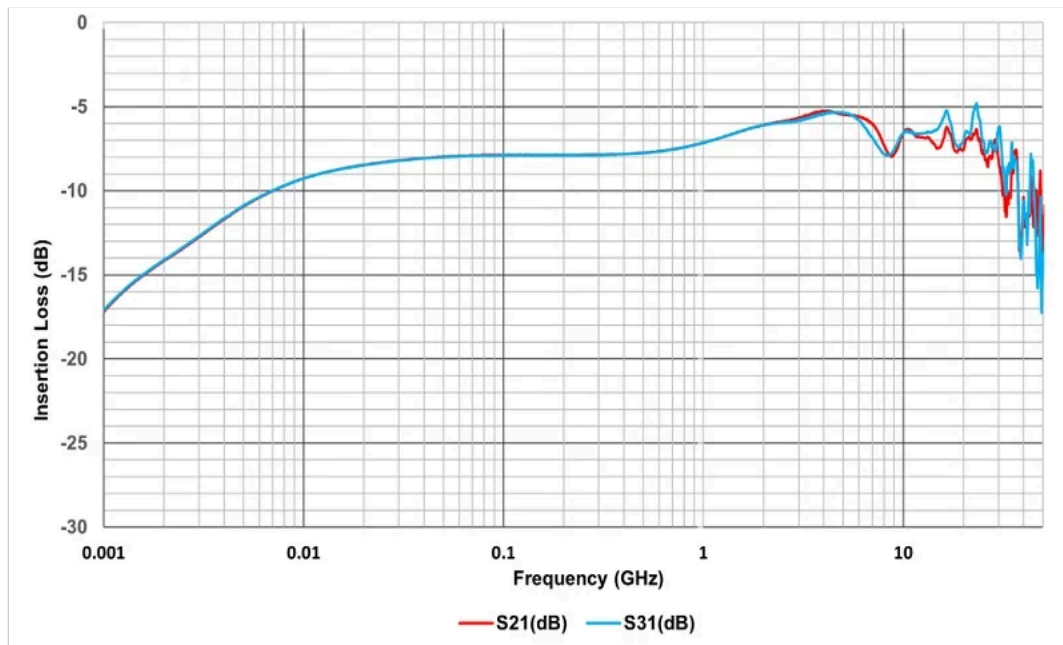


Figure 2. S_{21} and S_{31} vs. frequency for the HL9492. Image used courtesy of [Hyperlabs](#)

Because the input power is split equally between the two outputs, the insertion loss should theoretically be -3 dB. However, any real-world balun implementation will involve loss mechanisms that further reduce the power transmitted to the balanced outputs, leading to insertion loss values more negative than -3 dB. The magnitude of this loss depends on the particulars of the balun design.

There are several different methods of implementing baluns that can affect the overall shape of the frequency response. For example, Figure 3 shows the simulated frequency response of a transmission line balun constructed with coaxial cables. In this case, a phenomenon known as the half-wavelength resonance sets the upper limit for the usable bandwidth.

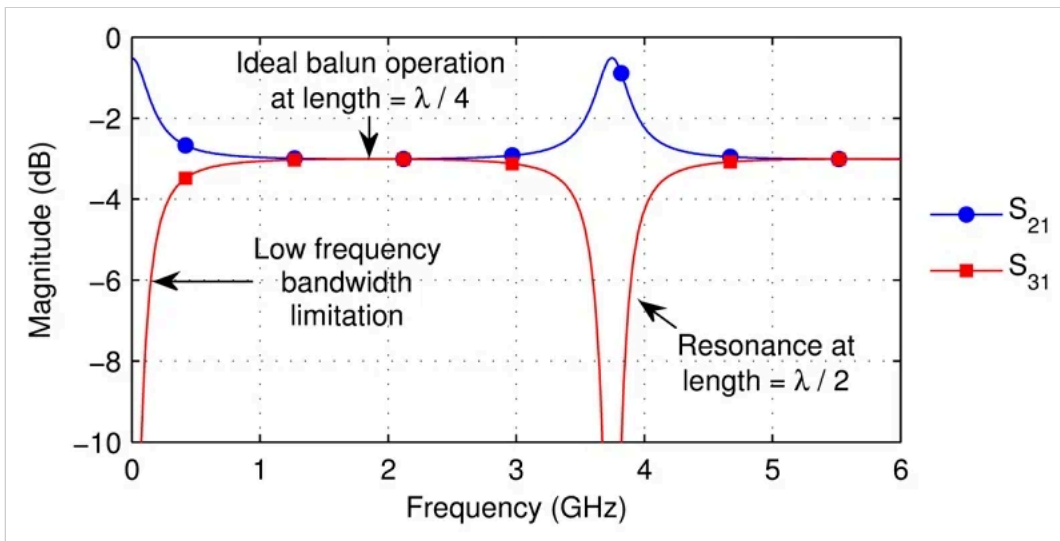


Figure 3. Half-wavelength resonance in a coaxial cable transmission line balun. Image used courtesy of [Robert M. Smith](#)

Return Loss

Return loss is the loss that the incident signal experiences as it reflects, or returns, from the balun's ports. Figure 4 shows the single-ended return loss of the HL9492.

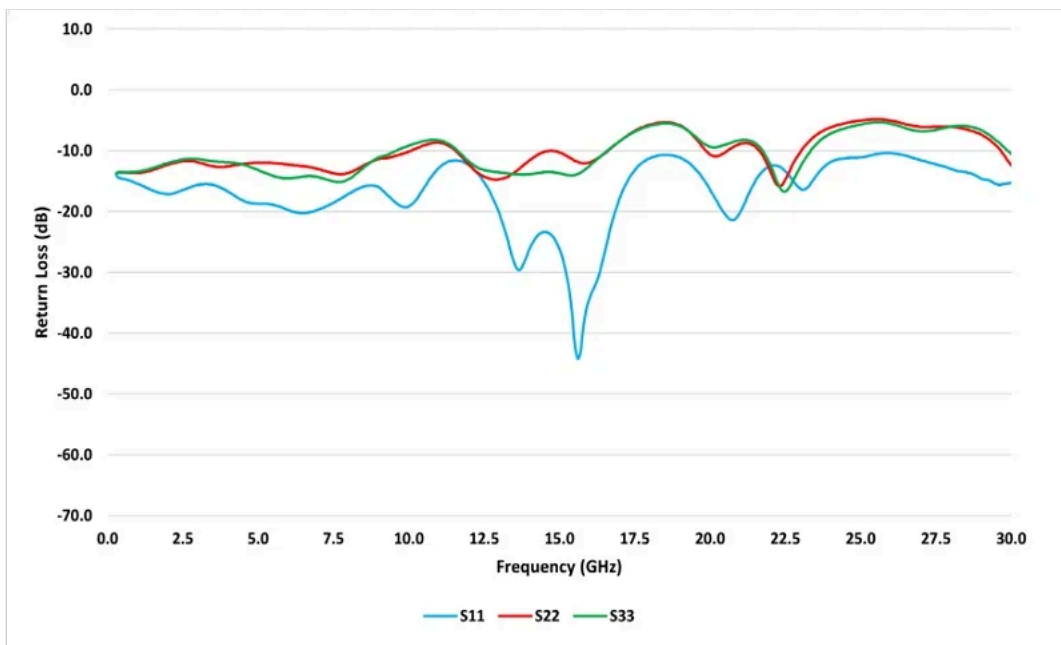


Figure 4. Single-ended return loss vs. frequency for the HL9492. Image used courtesy of [Hyperlabs](#)

When the insertion loss is low and the input return loss is high, the device can transfer a larger portion of the input power to the output. This provides us with a larger dynamic range.

In Figure 4, the return losses of Ports 2 and 3 are characterized individually. We could also usefully characterize both Ports 2 and 3 as a single, balanced port, as we did in our discussion of Figure 1. This model, shown in Figure 5, allows us to appropriately terminate the unbalanced port (Port 1) and apply a differential signal to the balanced port.

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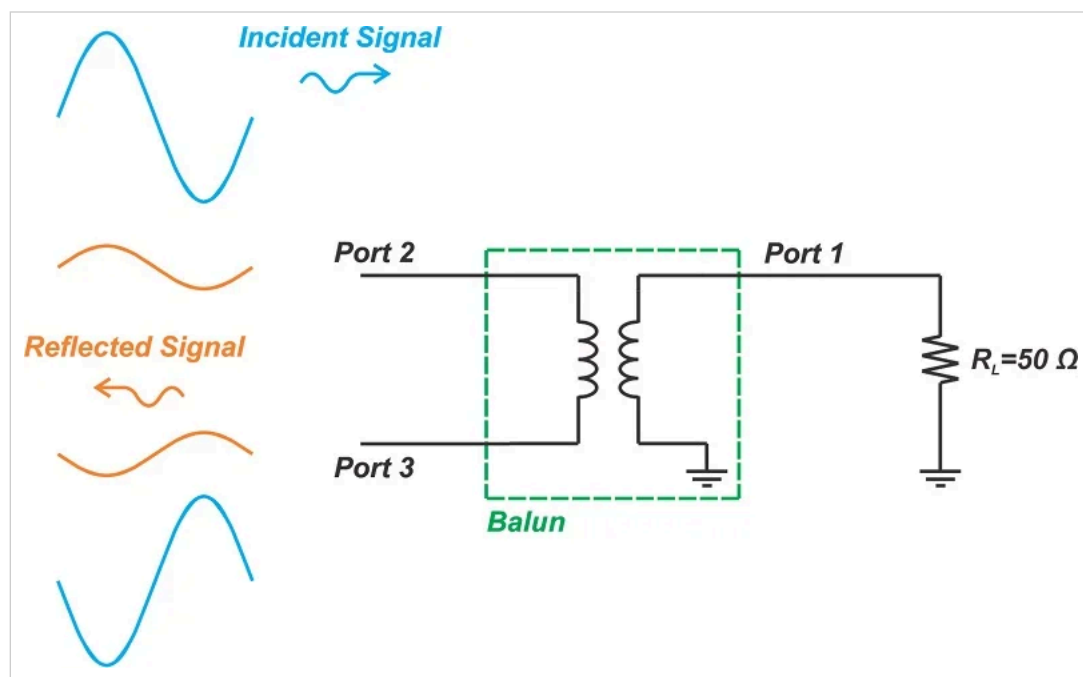


Figure 5. Characterizing return loss of Ports 2 and 3 as if they were a single, balanced port. Image used courtesy of Steve Arar

Ideally, a differential signal should pass through the balun completely, leading to a return loss of $-\infty$. However, as illustrated above, a practical balun reflects a small portion of the incident signal. Figure 6 shows the balanced output return loss of the Macom [MABA-011131](#) balun.

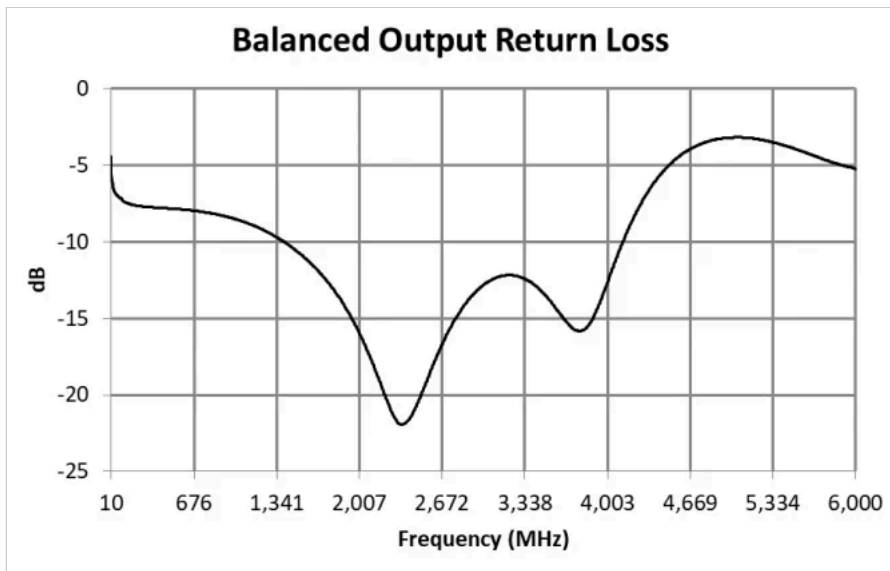


Figure 6. Balanced output return loss of the MABA-011131. Image used courtesy of [Macom](#).

A balanced signal incident on the balanced port is mostly absorbed, but a common-mode signal incident on the balanced port is mostly reflected. Ideally, the return loss of the balanced port for a common-mode signal is 0 dB. This is illustrated in Figure 7.

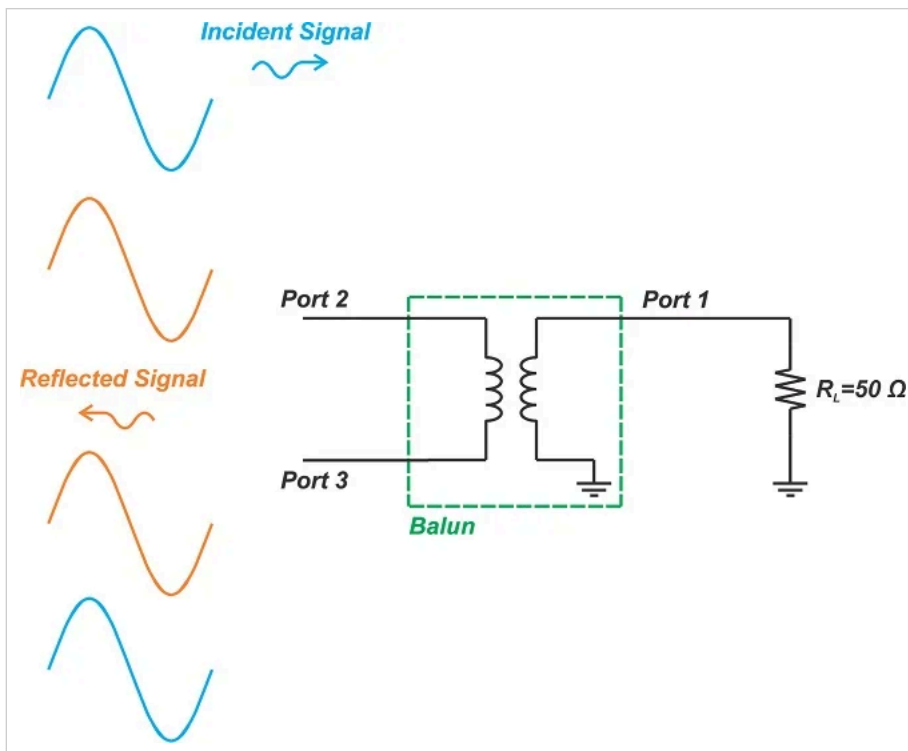


Figure 7. Most of the common-mode signal incident on the balanced port is reflected back. Image used courtesy of Steve Arar

It's worthwhile to mention that a practical balun may exhibit mode conversion. When applying a differential signal to the balanced port, we might observe a small common-mode signal being reflected from the device. The application of a common-mode signal might also produce a small, mode-converted, differential-mode signal bouncing back from the device.

These mode conversion effects are normally assumed to be negligible, and so details about them aren't included in most datasheets. The datasheet for the MABA-011131 balun we looked at above, for example, only provides the balanced insertion loss for the balanced port.

Amplitude and Phase Imbalance

The amplitude and phase imbalance parameters measure how well a balun converts a single-ended signal to a differential signal, or vice versa. They're perhaps the most important performance parameters for a balun, and deserve a more thorough explanation than we have time for in this article. For now, we'll keep the explanations brief.

The amplitude balance characterizes the match between the power magnitudes of the balanced ports. The amplitude *imbalance* is equal to the difference in magnitude between the two insertion loss terms (S_{21} and S_{31}). Ideally, the output power should be equal at both ports, giving us an amplitude imbalance of zero. In reality, however, there's always some mismatch due to the balun's design and fabrication.

Likewise, while the output signals should ideally be 180 degrees out of phase with one another, there's always some deviation due to the imperfections of practical baluns. The phase angle's deviation from the ideal 180 degrees is called phase imbalance.

Low-performance baluns typically have an amplitude imbalance of ± 1 dB and phase imbalance of ± 10 degrees. Higher-performance baluns, however, will have amplitude and phase imbalance values as small as ± 0.2 dB and ± 2 degrees, respectively.

Common-Mode Gain and Rejection Ratio

As mentioned above, a common-mode signal incident on the balanced port is, ideally, completely reflected. In practice, some of the input common-mode power gets absorbed, producing an unwanted signal at the single-ended output. As the device is reciprocal, this also means that power can scatter from the unbalanced port to the balanced output. We can quantify this effect by using the following equation to calculate the common-mode gain of the balun:

$$G_{cm} = \frac{S_{21} + S_{31}}{\sqrt{2}}$$

Equation 4.

The concept of the common-mode rejection ratio (*CMRR*), [adapted from low-frequency analog design](#), can now be applied. *CMRR* characterizes how well the device attenuates common-mode signals while producing the desired differential signal. Equations 3 and 4 lead to:

$$CMRR = \left| \frac{G_{dm}}{G_{cm}} \right| = \left| \frac{S_{21} - S_{31}}{S_{21} + S_{31}} \right|$$

Equation 5.

Let's cement these concepts by looking at an example.

Calculating the CMRR of a Balun

Assume that, at a given frequency, the transmission characteristics of a balun in terms of its conventional S-parameters are $S_{21} = 0.66 \angle 0$ degrees and $S_{31} = 0.75 \angle -170$ degrees. Let's calculate the differential-mode gain, common-mode gain, and *CMRR* of this balun.

First, we'll find the phase imbalance and amplitude imbalance. We can see from the above S-parameters that the device has 10 degrees of deviation from the ideal 180-degree phase angle, giving us our phase imbalance. Converting those S-parameters to decibel values, we see that $|S_{21}| = -3.61$ dB and $|S_{31}| = -2.5$ dB. These values correspond to an amplitude imbalance of 1.11 dB.

Plugging the linear forms of the S-parameters into Equations 3 and 4 results in $G_{dm} = -0.06$ dB and $G_{cm} = -19.4$ dB, respectively. Using either these gain values or the original S-parameters in Equation 5, we find that the *CMRR* is equal to 19.3 dB.

A high *CMRR* is directly related to good amplitude and phase balance characteristics. The example we worked through represents a typical low-performance balun with an amplitude imbalance of ± 1 dB and a phase imbalance of ± 10 degrees. As we saw, this balun can provide a *CMRR* of about 20 dB.

Wrapping Up

In this article, we learned about the basic concepts and most important performance parameters of baluns, which are essential to many radio frequency systems. I hope our discussion has enhanced your understanding of these important devices.

This article is Part 2 in a series on RF transformers and baluns. A complete list of articles in this series can be found below.

1. [Understanding the Non-Idealities of Magnetically Coupled RF Transformers](#)
2. Balun Basics and Practical Performance Parameters
3. [Introduction to Transmission Line Transformers and the Bifilar Coil](#)
4. [Understanding the Guanella Transmission Line Balun](#)
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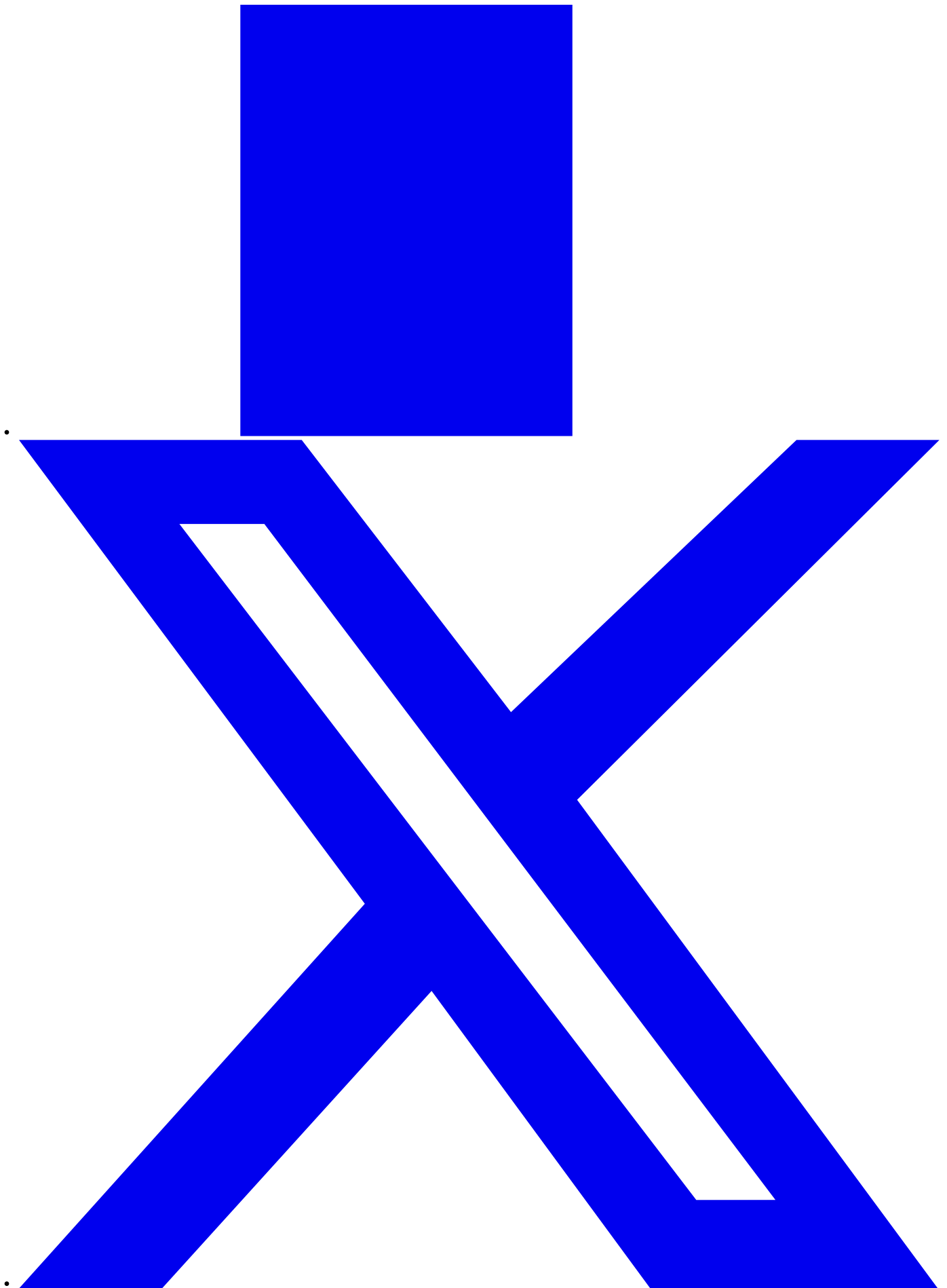
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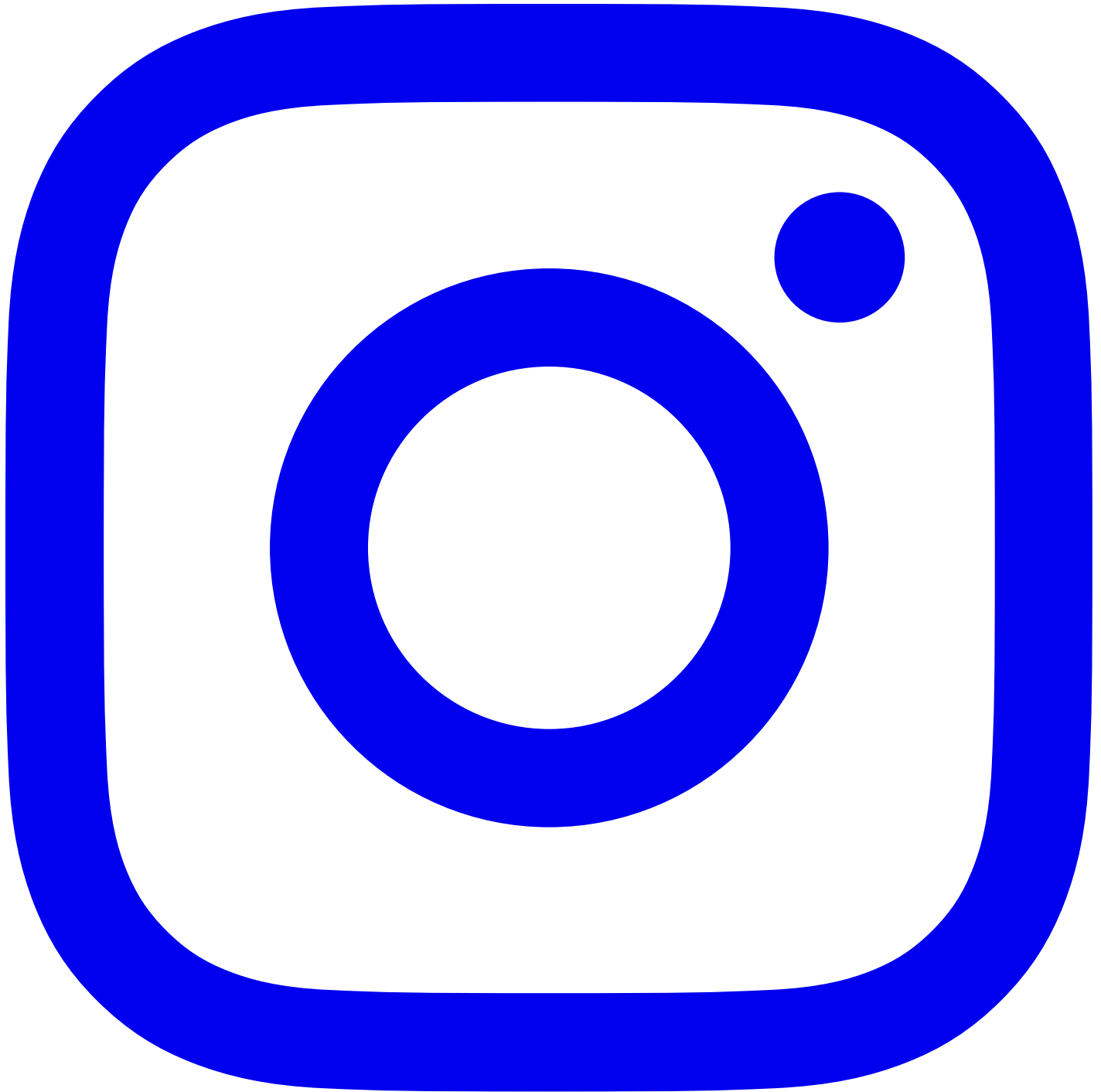
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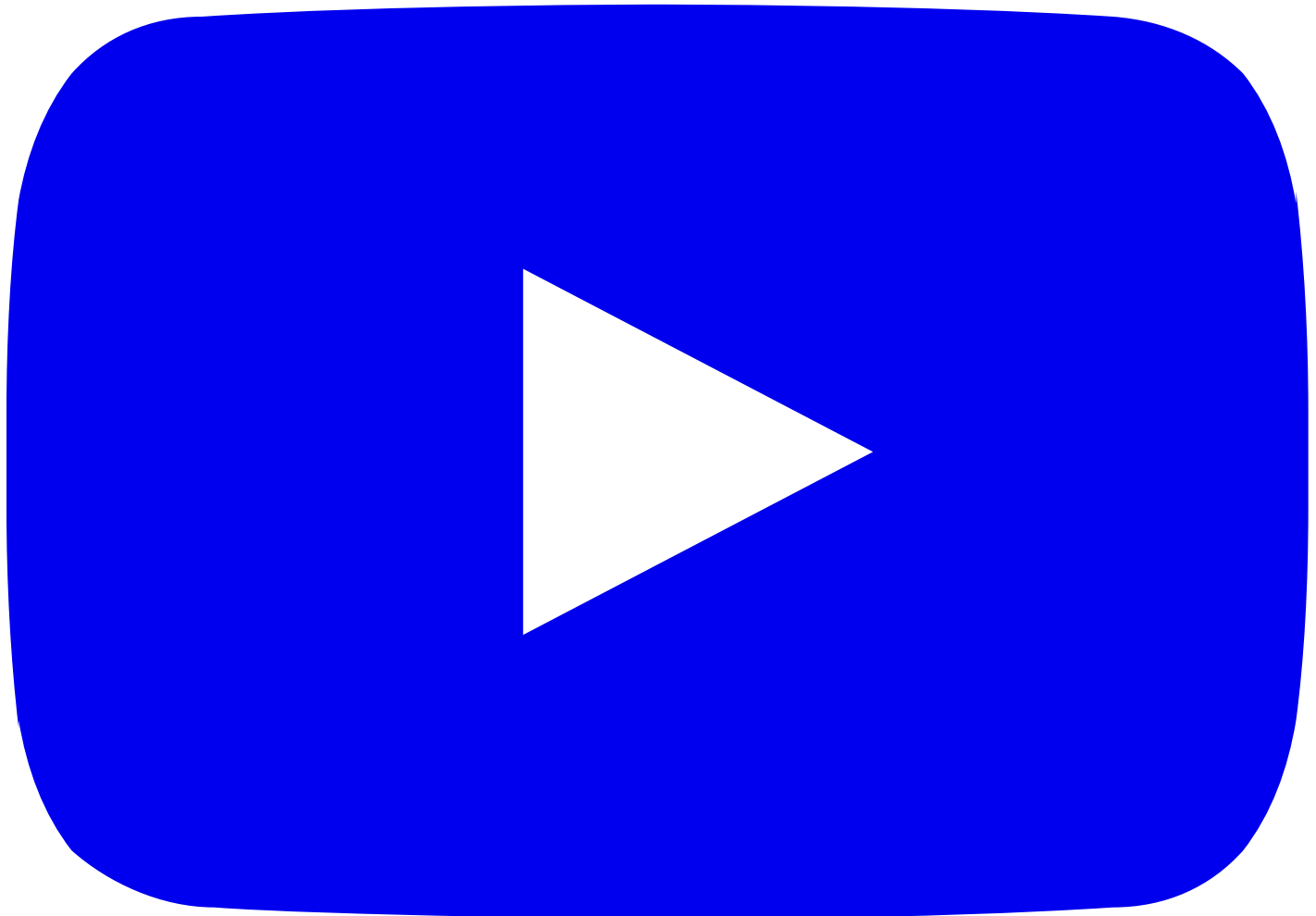
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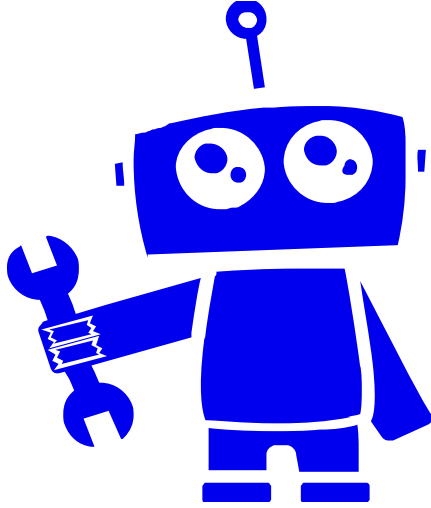
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